MILITARY SPECIFICATION

TESTER, AIRFLOW, GAS-PARTICULATE

FILTER UNIT, M39

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of airflow tester for measuring airflow through hoses of gas particulate filter units with airflow rates up to 5 cubic feet per minute.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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FSC 6680

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-77-2120 - Tester, Airflow, Gas-Particulate Filter Unit, M39.
PS 5-77-2120 - Tester, Airlfow, Gas-Particulate Filter Unit, M39.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurementt functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-77-2120 or on the subsidiary drawings shall conform to the specifications listed theron, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components shall conform to the specifications and drawings listed on TDPL 5-77-2120 and subsidiary drawings.

3.2 Assembly. The tester shall be assembled as specified on Drawing E5-77-2120.

3.3 Performance. The tester shall satisfy the pressure versus flow requirements of table I when tested as specified in 4.3.1. Prior to test, the pressure shall be calibrated in accordance with 4.3.3.

3.4 Adhesion strength. After assembly and cure the alternate design (adhesive bonded) orifice ring when tested in accordance with 4.3.2 shall show no evidence of separation from the tube when subjected to a force of 50 pounds applied axially for a minimum of 5 seconds, from the end opposite the ring tending to force the ring from the recess in the tube.

3.5 Workmanship. The tester shall be free of defects adversely affecting its appearance or serviceability. All welding, brazing, riveting, identifying of parts, and finish shall exhibit high quality workmanship. The unit shall be clean, free of material chips, and all foreign materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection

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the micrometer at 0 second for both the horizontal and vertical flights

TABLE I. Performance

Air Pressure		Airflow	
(Inches of Water)	Pascals	(SCFM)	Liter/Sec
0.7 ± 0.1	(174)	1.8 ± .15	(0.93 ± 0.07)
1.0 ± 0.12	(249)	2.2 ± .15	(1.04 ± 0.07)
2.0 ± 0.12	(498)	3.0 ± .15	(1.42 ± 0.07)
3.0 ± 0.12	(746)	3.6 ± .15	(1.70 ± 0.07)
4.0 ± 0.12	(995)	4.2 ± .15	(1.98 ± 0.07)
5.0 ± 0.12	(1244)	4.6 ± .15	(2.17 ± 0.07)

NOTE: SCFM denotes standard cubic feet per minute (70°F, 29.92 in Hg, 0.0750 lb/ft^3).
Metric equivalents are given for general information only.

requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inpsection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the testers produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.2.2 Sampling.

4.2.2.1 Examination and nondestructive tests. Sampling for examination shall be in accordance with MIL-STD-105. Each tester in the lot shall be examined for marking and serial number and tested for performance.

4.2.2.2 Destructive tests. Sample calibrated orifice assemblies shall be taken at random from the lot in accordance with MIL-STD-105, level S-4 for testing for adhesion strength.

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4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive test. Sample testers shall be examined in accordance with the classification of defects and MIL-STD-105. Each tester shall be examined for marking and serial number, and tested for performance (4.3.1). Any tester found defective shall be rejected.

4.2.3.2 For destructive test. Sample calibrated orifice assemblies shall be tested for adhesion strength in accordance with 4.3.2, the acceptance number is zero. No calibrated orifice assembly selected for the adhesion strength test shall be put into production or delivered as part of a production quantity.

4.2.3.3 Classification of defects.

(a) Tester, airflow, gas-particulate filter unit M39 (Dwg E5-77-2120).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100% inspection	
101	Performance	4.3.1
102	Pressure gage calibration	
103	Marking and serial number missing or incorrect	
	AQL 1.0 percent defective	
104	Components missing or incorrectly assembled	
105	Clip assemblies and bracket locations incorrect	
106	Instruction card legibility or contents incomplete	
107	Protective finish missing or color incorrect	
108	Workmanship (3.5)	

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(b) Calibrated orifice assembly (Dwg D5-77-2115).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimension (0.187) incorrect
102	Dimension (0.080) incorrect
103	Dimension (0.492) incorrect
104	Angle (53°) incorrect
105	Angle (6°) incorrect
106	Dimension (0.109) incorrect
107	Height (0.734) incorrect
108	Dimension (0.094) incorrect
109	Diameter (0.865) incorrect
110	Diameter (0.723) incorrect
111	Machine finish 63 incorrect
112	Burrs
113	Diameter (0.765) incorrect
Minor:	AQL 2.5 percent defective
201	Dimension (0.110) incorrect
202	Dimension (0.056) incorrect
203	Dimension (0.024) incorrect
204	Angles (45°) incorrect
205	Dimension (0.75) incorrect
206	Overall length incorrect
207	Diameter (0.641) incorrect

(c) Adapter connector (Dwg C5-77-2109).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Components missing or incorrectly assembled
102	Workmanship (3.5)

4.3 Tests.

4.3.1 Performance. The tester shall be connected in series with a suitable test device consisting of a dry gas meter, pressure measuring device and a regulated air source. The accuracy of the gas meter shall be ± 2% and the pressure measuring device ± 0.5% full scale. The air source shall be regulated until

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the dry gas meter indicates the airflow values specified in table I. The corresponding air pressure values shall be checked for compliance with 3.3.

4.3.2 Adhesion strength test. Calibrated orifice assemblies shall be tested to the requirements of 3.4 with a suitable spring or dead weight force measuring device.

4.3.3 Pressure gage calibration. The pressure gage shall be calibrated in accordance with calibration procedure TB 3-6685-100-50.

5. PACKAGING

5.1 Preservation, unit packing, packing, and marking. Preservation, unit packing, packing, and marking for all levels shall be accomplished in accordance with the Packaging Data Stock Number or its "PS" part number (PS 5-77-2120).

6. NOTES

6.1 Intended use. The airflow tester covered by this specification is intended for use in measuring the airflow through the output air hoses of the following types of gas-particulate filter units:

	Type
ABC-M7A1	Hospital Six, Man, 12 CFM
M8 Series	Tank, 12 CFM
M13	Tank, 20 CFM
M14	Armored Ambulance, 12 CFM

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1).
(c) Applicable serial numbers.

6.3 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-T-1040 dated 17 September 1969.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6680-A183

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